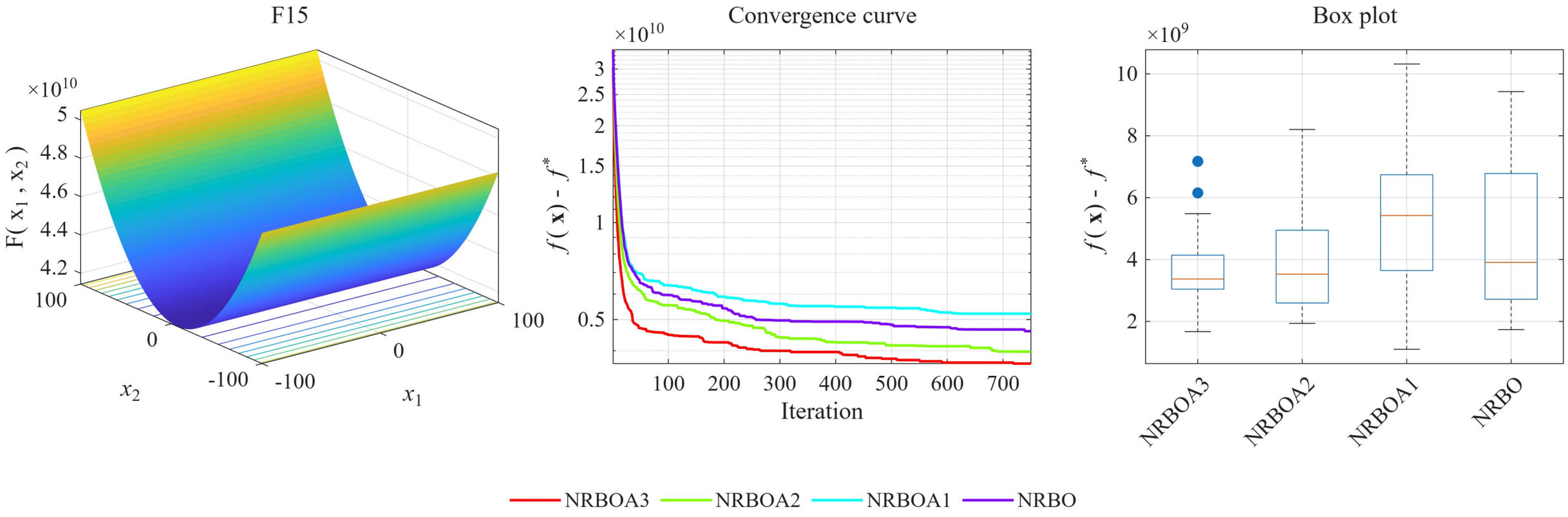

F15
×10^10
5
4.8
4.6
4.4
4.2
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
×10^10
3
2.5
2
1.5
1
0.5
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^9
10
8
6
4
2
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO